A uniform quadratic wedge for consecutive Toeplitz minors of the Riemann ξ-coefficients

Alessio Doria – Independent Researcher

numero2405@gmail.com

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Abstract. Let $\frac{1}{8}\xi(\frac{1}{2}+\frac{ix}{2}) = \sum_{k\ge 0}(-1)^k a_k x^{2k}$, so that $a_k > 0$, and let $D_{r,k} = \det[a_{k+j-i}]_{i,j=0}^{r-1}$ be the consecutive Toeplitz minors of the coefficient sequence. The Riemann Hypothesis implies that (a_k) is a Pólya frequency sequence, hence $D_{r,k} \ge 0$ for all r, k. Michałowski recently proved unconditionally that $D_{r,k} > 0$ whenever $r \ge 2$ and $k \ge 10^{18} r^3$. We prove that

$$D_{r,k} > 0 \qquad \text{for all } r \ge 2 \text{ and } k \ge 10^{11} r^2.$$

The improvement comes from replacing the three-point log-quadratic comparison sequence by a four-point q-Pochhammer comparison sequence $c_s = (1-\alpha)^s/(\alpha;\mathfrak{q})_s$ that reproduces the normalized coefficients at four consecutive indices exactly. The reversed Hankel block of this reference is diagonally congruent to a matrix with an exact q-Pascal LDL^T factorization (a consequence of the q-Chu–Vandermonde summation) with alternating pivots, so the dilation/whitening argument applies to it unchanged, while the logarithmic mismatch now has four interpolation zeros. The weighted remainder therefore starts at fourth order and scales like r^2/k instead of $(r^3/k)^{1/2}$. The only analytic inputs are Michałowski's all-degree Taylor bound for $\log a$ and the Liu–Mao asymptotic for the log-curvature of (a_k); all remaining constants are certified in interval arithmetic.

Keywords. Riemann ξ-function; Toeplitz minors; Pólya frequency sequences; total positivity; q-Pochhammer symbol; q-Chu–Vandermonde; Sylvester inertia.

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1. Introduction

1.1. Setting

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and write, following [10],

$$\Xi(x) = \frac{1}{8}\,\xi\Big(\frac{1}{2}+\frac{ix}{2}\Big) = \sum_{k\ge 0}(-1)^k a_k x^{2k}, \qquad a_k = \frac{1}{(2k)!}\int_0^\infty u^{2k}\Phi(u)\,du > 0, \tag{1.1}$$

where $\Phi(u) = \sum_{n\ge 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})\exp(-\pi n^2 e^{4u}) > 0$. For $r \ge 1$ and $k \ge 0$ set

$$D_{r,k} = \det\left[a_{k+j-i}\right]_{i,j=0}^{r-1} \qquad (a_\ell = 0 \text{ for } \ell < 0).$$

The function $G(z) = \sum_k a_k z^k$ is entire of order $1/2$ and $\Xi(x) = G(-x^2)$. The Riemann Hypothesis (RH) is equivalent to all zeros of G being real and negative, and by the Aissen–Schoenberg–Whitney–Edrei theorem [1, 3] this happens if and only if (a_k) is a Pólya frequency sequence, i.e. every minor of the Toeplitz matrix $[a_{j-i}]_{i,j\geq 0}$ is nonnegative. The consecutive minors $D_{r,k}$ form a natural two-parameter subfamily; under RH they are all nonnegative.

Unconditional information on $D_{r,k}$ is of several kinds. For orders two and three positivity is known for every k: $D_{2,k} > 0$ is the Turán inequality of Csordas, Norfolk and Varga [2], and $D_{3,k} > 0$ is due to Nuttall [11] (see [13]). For each fixed r, Katkova [8] proved that $D_{r,k} > 0$ for $k \geq N(r)$ with a non-explicit $N(r)$. Wang and Yang report an order-uniform sufficient range of exponential size, $k > c\, e^{r(r-1)}$, for the same determinants [13, Introduction]. More recently Michałowski [10] obtained the first explicit polynomial uniform wedge,

$$D_{r,k} > 0 \qquad (r \geq 2,\ k \geq 10^{18} r^3). \tag{1.2}$$

Several related results concern different hierarchies of the Laguerre–Pólya program. Wang and Yang [13] prove Laguerre inequalities of order r for $k > cr^3$ and complete-monotonicity estimates for the Ξ-coefficients; Griffin–Ono–Rolen–Zagier [5] and Holland [6] establish hyperbolicity regions for Jensen polynomials of ξ; and Liu and Mao [9] prove that (a_k) (equivalently $\lambda_n = \xi^{(2n)}(\frac{1}{2})/(2n)!$) is strictly infinitely log-concave. Sra [12] proves log-concavity and Pólya frequency properties of large finite order for the densities whose Laplace transforms are $1/\xi(\frac{1}{2} + s)$ and $1/\Xi$, which are different objects from the coefficient sequence (a_k). These conditions do not imply positivity of the order-r consecutive Toeplitz minors considered here. To the best of our knowledge, the result below is the first polynomial uniform wedge for these minors with exponent strictly below three.

1.2. Result

Main Theorem. *For every integer $r \geq 2$ and every integer $k \geq 10^{11} r^2$ one has $D_{r,k} > 0$.*

The constant 10^{11} is not optimized; the argument closes for every $C > 1.35 \cdot 10^{10}$ (Remark 8.2). The theorem says nothing about the RH-critical regime $k = O(r)$.

1.3. Strategy

We follow the architecture of [10]: compare the reversed Hankel block $H = [a_{k+d-i-j}]_{i,j=0}^{d}$, $d = r - 1$, with an explicit reference block $M = SJS^T$ of known inertia, and show that the *whitened* perturbation $S^{-1}(H' - M)S^{-T}$ has spectral norm < 1, where H' is a diagonal rescaling of H; Sylvester's law then fixes the sign of $\det H$. The new ingredient is the reference. In [10] the reference is $c_s = q_k^{s(s-1)/2}$, which matches $\log a_{k+s}$ at the three points $s = -1, 0, 1$, so the logarithmic error h has a cubic Taylor term of size k^{-2}. Here we use the q-Pochhammer sequence

$$c_s = \frac{(1-\alpha)^s}{(\alpha;\mathfrak{q})_s},$$

whose two parameters are chosen so that c matches the normalized coefficients at $s = -1, 0, 1, 2$. LDL^T factorizations of q-Pascal type are classical; what is new here is their use for a four-point reference adapted to the ξ-coefficients. Two facts make this work:

1. the reversed Hankel block of c is diagonally congruent to the matrix

$$U_{ij} = \frac{(B;Q)_{i+j}}{(B;Q)_i\,(B;Q)_j},$$

which has an exact LDL^T factorization with the *same* q-Pascal factor $L = [\begin{bmatrix} i \\ j \end{bmatrix}_Q]$ as the q-Vandermonde matrix and with pivots of alternating sign (Proposition 5.2);

2. the response of the whitened block to a diagonal dilation depends only on L and on the pivot ratios, and the new pivot ratios give a whitened sub-diagonal that is no larger than before (Lemma 6.2).

Since the error h now vanishes at four nodes, it equals $\sum_{n\ge 4}\beta_n(s^n - \mathcal{I}_3[s^n])$ with $|\beta_n| \lesssim 40^n k^{1-n}$, and the relevant weight is $w \asymp \sqrt{rk}$; the leading term is $k^{-3}w^4 \asymp r^2/k$.

1.4. Inputs and logical dependencies

We use exactly two analytic results from the literature, stated in Section 2: the k-unit form of Michałowski's all-degree bound ([10, Prop. 3.7, first display]) and the Liu–Mao estimate [9, Lemma 3.3]. We do *not* use [10, Lemma 2.1] (the bounds $1/(2k) < \tau_k < 4/k$) nor its τ-unit form of Gate A: all information on τ_k comes from Input B. The dilation estimates and the inertia lemma are re-proved here for the new reference. All scalar inequalities are certified with Arb [7]; scripts are provided as ancillary files.

Notation

$(a;q)_n = \prod_{j=0}^{n-1}(1-aq^j)$ for $n \ge 0$, $(a;q)_{-m} = 1/(aq^{-m};q)_m$ for $m > 0$, and $\begin{bmatrix} n \\ m \end{bmatrix}_Q = (Q;Q)_n/((Q;Q)_m(Q;Q)_{n-m})$. Throughout $r \ge 2$, $d = r-1$, and $\|\cdot\|$ is the spectral norm.

2. Analytic inputs

Put $f = \log a$, where $a(z) = I(z)/\Gamma(2z+1)$ and $I(z) = \int_0^\infty u^{2z}\Phi(u)\,du$ is holomorphic on $\operatorname{Re} z > -\frac{1}{2}$; thus $a(n) = a_n$ for integers $n \ge 0$.

Input A (Michałowski, Gate A) ([10, Lemma 3.6, Prop. 3.7])**.** *For every integer $k \ge 10^9$, f is holomorphic on $|z-k| \le 0.05k$ (with $f(n) = \log a_n$ at integers), and*

$$\Big|\frac{f^{(n)}(k)}{n!}\Big| \le 3\cdot 40^n\, k^{1-n} \qquad (n \ge 3). \tag{2.1}$$

Define the log-curvature

$$q_k = \frac{a_{k-1}a_{k+1}}{a_k^2} = e^{-\tau_k}, \qquad \tau_k > 0.$$

Liu and Mao use $F(x) = \xi(\frac{1}{2} + \sqrt{x}) = \sum \lambda_n x^n$. Since $\frac{1}{8}\xi(\frac{1}{2} + \frac{\sqrt{z}}{2}) = \sum a_n z^n = \frac{1}{8}F(z/4)$, we have $a_n = \lambda_n/(8\cdot 4^n)$, and therefore $\lambda_{n-1}\lambda_{n+1}/\lambda_n^2 = q_n$. Let W be the principal branch of the Lambert function and

$$\mathcal{T}(z) = \frac{2\,W(2z/\pi)}{z\,(1+W(2z/\pi))}.$$

Input B (Liu–Mao, Lemma 3.3) ([9, Lemma 3.3, (3.53)]). *For real $n \geq 12000$ there is a function g_0, holomorphic on $D_0 = \{z : |z - n| \leq n/2\}$ and real on $D_0 \cap \mathbb{R}$, with $g_0(m) = \log(\lambda_m^2/\lambda_{m-1}\lambda_{m+1}) = \tau_m$ for integers $m \in D_0$, and*

$$|g_0(z) - \mathcal{T}(z)| \leq \frac{20}{n^2} \qquad (z \in D_0).$$

Remark 2.1. Input A is used only through the Taylor coefficients of f at an integer centre; any analytic interpolation of $\log a_n$ with these bounds would do, and it need not coincide with the interpolation g_0 of Input B, since only the values at integers enter the comparison (Section 7).

3. Curvature windows

Lemma 3.1. *For every integer $k \geq 12000$,*

$$1.71 < \frac{12}{7} - \frac{20}{k} \leq k\tau_k \leq 2 + \frac{20}{k} < 2.002.$$

Proof. Write $w = W(2x/\pi)$. Since $6e^6 < 2 \cdot 12000/\pi$ and W is increasing, $w > 6$ for $x \geq 12000$, so $x\mathcal{T}(x) = 2w/(1+w) \in [12/7, 2)$. Apply Input B with $n = k$ at $z = k$. □

Lemma 3.2. *For every integer $k \geq 12000$, $\Delta_k := \tau_k - \tau_{k+1}$ satisfies*

$$1.675 < \frac{576}{343}\Big(\frac{k}{k+1}\Big)^2 - \frac{40}{k-2} \leq k^2\Delta_k \leq 2 + \frac{40}{k-2} < 2.004.$$

In particular $q_k < q_{k+1}$.

Proof. With $w = W(2x/\pi)$ and $w'(x) = w/(x(1+w))$ one computes

$$-\mathcal{T}'(x) = \frac{2}{x^2}\, g(w), \qquad g(w) = \frac{w^2(w+2)}{(1+w)^3}.$$

Since $(\log g)'(w) = \frac{2}{w} + \frac{1}{w+2} - \frac{3}{w+1} = \frac{w+4}{w(w+1)(w+2)} > 0$, g increases from $g(6) = 288/343$ to 1; hence $\frac{576}{343}x^{-2} \leq -\mathcal{T}'(x) < 2x^{-2}$ for $x \geq 12000$, and $\mathcal{T}(k) - \mathcal{T}(k+1) \in [\frac{576}{343}(k+1)^{-2},\, 2k^{-2}]$. Let $E = g_0 - \mathcal{T}$ with $n = k$ in Input B. For $\eta \in [k, k+1]$ the disc $|z - \eta| \leq k/2 - 1$ lies in D_0, so Cauchy's estimate gives $|E'(\eta)| \leq (20/k^2)/(k/2-1) = 40/(k^2(k-2))$, whence $|E(k) - E(k+1)| \leq 40/(k^2(k-2))$. Both bounds are monotone in k in the favourable direction; the numerical values at $k = 12000$ are certified in Appendix A. □

4. The four-point q-Pochhammer reference

Fix an integer $k \geq 12000$ and normalize

$$\rho = \frac{a_{k+1}}{a_k}, \qquad b_s = \frac{a_{k+s}}{a_k\, \rho^s},$$

so that $b_{-1} = q_k =: u$, $b_0 = b_1 = 1$, $b_2 = q_{k+1} =: v$, and $0 < u < v < 1$ by Lemmas 3.1–3.2.

Definition 4.1. Let

$$\mathfrak{q} = \frac{1-v}{v(1-u)}, \qquad \alpha = \frac{(1-u)(1-v)}{v-u}, \qquad \theta = \log \mathfrak{q}, \qquad c_s = \frac{(1-\alpha)^s}{(\alpha;\mathfrak{q})_s} \quad (s \in \mathbb{Z}).$$

Lemma 4.2. *$c_{-1} = u$, $c_0 = c_1 = 1$, $c_2 = v$. Moreover, with $\sigma = \tau_{k+1}$ and $\Delta = \tau_k - \sigma$,*

$$\alpha = \frac{2\sinh(\tau_k/2)\sinh(\sigma/2)}{\sinh(\Delta/2)}, \qquad \theta = \frac{\tau_k + \sigma}{2} + \log\frac{\sinh(\sigma/2)}{\sinh(\tau_k/2)}.$$

Proof. $c_0 = 1$, $c_1 = 1$, $c_2 = (1-\alpha)/(1-\alpha\mathfrak{q})$ and $c_{-1} = (1-\alpha\mathfrak{q}^{-1})/(1-\alpha)$. Substituting the definitions of $\alpha, \mathfrak{q}$ gives $c_2 = v$ and $c_{-1} = u$ after clearing denominators (a polynomial identity in u, v). The hyperbolic forms follow from $1 - e^{-x} = 2e^{-x/2}\sinh(x/2)$. □

Lemma 4.3. *For $k \geq 12000$: $1.45 < \alpha < 2.40$ and $0.53 < k\theta < 2.002$. In particular $\mathfrak{q} > 1$.*

Proof. Using $x \leq \sinh x \leq xe^{x^2/6}$, α lies between $\frac{\tau_k\sigma}{\Delta}e^{-\Delta^2/24}$ and $\frac{\tau_k\sigma}{\Delta}e^{(\tau_k^2+\sigma^2)/24}$. For θ: $(\tau_k + \sigma)/2 = \sigma + \Delta/2 \geq \sigma$, $\log(\sigma/\tau_k) = -\log(1 + \Delta/\sigma) \geq -\Delta/\sigma$, and $\log\frac{\sinh(\sigma/2)}{\sinh(\tau_k/2)} \geq \log\frac{\sigma}{\tau_k} - \frac{\tau_k^2}{24}$; hence $\theta \geq \sigma - \Delta/\sigma - \tau_k^2/24$. Also $\theta < \tau_k$ because sinh is increasing. Insert Lemmas 3.1–3.2 and $k\sigma \geq k\tau_k - k\Delta_k$; see Appendix A. □

Lemma 4.4 (Positivity)**.** *If $(r-1)\theta < \log\alpha$, then $c_s > 0$ for $|s| \leq r-1$, and every factor $1 - \alpha\mathfrak{q}^j$ with $-(r-1) \leq j \leq r-2$ is negative.*

Proof. $\alpha\mathfrak{q}^j \geq \alpha e^{-(r-1)\theta} > 1$ for $j \geq -(r-1)$. For $s > 0$ numerator and denominator of c_s both have sign $(-1)^s$; for $s = -m < 0$, $c_s = (1-\alpha)^{-m}\prod_{j=1}^{m}(1-\alpha\mathfrak{q}^{-j})$ has $2m$ negative factors. □

Remark 4.5. The condition $1 < \mathfrak{q} < \alpha$ alone does *not* suffice for positivity at negative indices; the hypothesis $(r-1)\theta < \log\alpha$ is what is needed, and it holds in the wedge of the Main Theorem with enormous room (Appendix A).

4.1. Analytic continuation and Taylor coefficients

Put $z_0 = 1/\alpha \in (0.41, 0.69)$ and $Q = \mathfrak{q}^{-1} = e^{-\theta}$. Writing $1 - \alpha\mathfrak{q}^j = -\alpha\mathfrak{q}^j(1 - z_0Q^j)$ gives $c_s = (1-z_0)^s Q^{s(s-1)/2}/(z_0;Q)_s$ for integers s (for $s < 0$ by the cocycle relation $(a;q)_{m+n} = (a;q)_m(aq^m;q)_n$, valid for all integers m, n). With $(z_0;Q)_s = (z_0;Q)_\infty/(z_0Q^s;Q)_\infty$ and $\log(x;Q)_\infty = -\sum_{m\geq1} x^m/(m(1-Q^m))$ we obtain, for $|s| < \theta^{-1}\log(1/z_0)$,

$$\ell(s) := s\log(1-z_0) - \frac{\theta}{2}s(s-1) + \sum_{m\geq1}\frac{z_0^m(1-e^{-m\theta s})}{m\,(1-e^{-m\theta})}, \tag{4.1}$$

a holomorphic function, real on the real axis, with $e^{\ell(s)} = c_s$ at integers s in its domain.

Lemma 4.6. *For $n \geq 3$, $|[s^n]\,\ell(s)| \leq \kappa^n/n + \kappa^{n-1}/(n(n-1))$ with $\kappa = \theta/(1-z_0)$. For $k \geq 12000$ this gives $|[s^n]\ell(s)| \leq (9/k)^{n-1}$.*

Proof. For $n \geq 3$ only the series contributes, and $[s^n](1 - e^{-m\theta s}) = -(-m\theta)^n/n!$, so

$$|[s^n]\ell| \leq \frac{\theta^n}{n!}\sum_{m\geq1}\frac{m^{n-1}z_0^m}{1-e^{-m\theta}} \leq \frac{\theta^n}{n!}\Big(\sum_m m^{n-1}z_0^m + \frac{1}{\theta}\sum_m m^{n-2}z_0^m\Big),$$

using $1/(1-e^{-x}) \leq 1 + 1/x$. Since $\sum_{m\geq1} m^p z^m = zA_p(z)/(1-z)^{p+1}$ with the Eulerian polynomial A_p (nonnegative coefficients, $A_p(1) = p!$), $\sum_m m^p z^m \leq p!/(1-z)^{p+1}$ for $0 < z < 1$, $p \geq 0$. This gives the first bound. By Lemma 4.3, $k\kappa = k\theta\,\alpha/(\alpha-1) < 2.002 \cdot 1.45/0.45 < 6.46 < 8.4$, and $8.4^n/(nk) + 8.4^{n-1}/(n(n-1)) < 9^{n-1}$ for $n \geq 3$, $k \geq 12000$. □

5. The reversed Hankel block and its q-Pascal factorization

Set $d = r - 1$ and

$$M_{ij} = c_{d-i-j}, \qquad 0 \le i, j \le d, \qquad Q = \mathfrak{q}^{-1}, \qquad B = \alpha\mathfrak{q}^{d-1}.$$

Lemma 5.1 (Congruence). *$M = c_d\, A\, U\, A$ with $A = \mathrm{diag}(A_i)$, $A_i = (B;Q)_i/(1-\alpha)^i$, and*

$$U_{ij} = \frac{(B;Q)_{i+j}}{(B;Q)_i\,(B;Q)_j}.$$

Under the hypothesis of Lemma 4.4, $c_d > 0$ and $A_i > 0$.

Proof. By the integer cocycle, $(\alpha;\mathfrak{q})_d = (\alpha;\mathfrak{q})_{d-n}(\alpha\mathfrak{q}^{d-n};\mathfrak{q})_n$ for every $0 \le n \le 2d$, including $d - n < 0$, and $(\alpha\mathfrak{q}^{d-n};\mathfrak{q})_n = \prod_{l=0}^{n-1}(1 - \alpha\mathfrak{q}^{d-1-l}) = (B;Q)_n$. Hence $c_{d-n} = c_d\,(B;Q)_n/(1-\alpha)^n$, which is the claim with $n = i + j$. Signs follow from Lemma 4.4: $1 - BQ^l = 1 - \alpha\mathfrak{q}^{d-1-l} < 0$ for $0 \le l \le d-1$, so $(B;Q)_i$ and $(1-\alpha)^i$ both have sign $(-1)^i$. □

Proposition 5.2 (q-Pascal LDL^T). *For indeterminates B, Q and all $i, j \ge 0$,*

$$U_{ij} = \sum_{m=0}^{\min(i,j)} \begin{bmatrix} i \\ m \end{bmatrix}_Q \begin{bmatrix} j \\ m \end{bmatrix}_Q D_m, \qquad D_m = B^m Q^{m(m-1)}\,\frac{(Q;Q)_m}{(B;Q)_m}.$$

Thus $U = LDL^T$ with $L_{ij} = \begin{bmatrix} i \\ j \end{bmatrix}_Q$ and $D = \mathrm{diag}(D_m)$. In our situation $0 < Q < 1$ and $BQ^{m-1} = \alpha\mathfrak{q}^{d-m} > 1$ for $1 \le m \le d$, so $\operatorname{sgn} D_m = (-1)^m$.

Proof. From $(Q;Q)_i/(Q;Q)_{i-m} = (-1)^m Q^{im-\binom{m}{2}}(Q^{-i};Q)_m$ we get

$$\begin{bmatrix} i \\ m \end{bmatrix}_Q \begin{bmatrix} j \\ m \end{bmatrix}_Q D_m = \frac{(Q^{-i};Q)_m (Q^{-j};Q)_m}{(Q;Q)_m (B;Q)_m}\,(BQ^{i+j})^m.$$

Hence the sum equals ${}_2\phi_1(Q^{-i}, Q^{-j}; B; Q, BQ^{i+j})$, which by the q-Chu–Vandermonde summation ${}_2\phi_1(q^{-n}, b; c; q, cq^n/b) = (c/b;q)_n/(c;q)_n$ [4, §1.5 and Appendix II], with $n = i$, $b = Q^{-j}$, $c = B$, equals $(BQ^j;Q)_i/(B;Q)_i = U_{ij}$. The sign of D_m: $(Q;Q)_m > 0$ and every factor $1 - BQ^l$, $0 \le l \le m-1$, of $(B;Q)_m$ is negative. □

Consequently

$$M = SJS^T, \qquad S = \sqrt{c_d}\; A\, L\, |D|^{1/2}, \qquad J = \mathrm{diag}\left((-1)^m\right)_{m=0}^{d}. \tag{5.1}$$

6. Dilation response of the whitened reference

For a function ϕ defined at the integers $|s| \le d$ write $M \circ \phi := [c_{d-i-j}\phi(d-i-j)]_{i,j}$ and define the *response*

$$\mathcal{R}(\phi) = \left\| S^{-1}(M \circ \phi)S^{-T} \right\|.$$

Since $M \circ \phi = c_d A(U \circ \phi)A$ and diagonal matrices commute with the Hadamard product structure, $\mathcal{R}(\phi) = \| |D|^{-1/2} L^{-1}(U \circ \phi)L^{-T}|D|^{-1/2} \|$; in particular A and c_d play no role.

Lemma 6.1. *Let $\Lambda_\beta = \mathrm{diag}(Q^{\beta i})_{i=0}^{d}$ and $R_\beta = L^{-1}\Lambda_\beta L$. Then R_1 is lower bidiagonal with*

$$(R_1)_{mm} = Q^m, \qquad (R_1)_{m,m-1} = -Q^{m-1}(1-Q^m).$$

Proof. We need $\Lambda_1 L = L R_1$, i.e. $Q^i \begin{bmatrix} i \\ j \end{bmatrix}_Q = Q^j \begin{bmatrix} i \\ j \end{bmatrix}_Q - Q^j(1-Q^{j+1}) \begin{bmatrix} i \\ j+1 \end{bmatrix}_Q$. This follows from the q-Pascal relation $(1-Q^{j+1}) \begin{bmatrix} i \\ j+1 \end{bmatrix}_Q = (1-Q^{i-j}) \begin{bmatrix} i \\ j \end{bmatrix}_Q$. □

Lemma 6.2 (Whitened sub-diagonal)**.** *Let $\widehat{R}_\beta = |D|^{-1/2} R_\beta |D|^{1/2}$. Then $(\widehat{R}_1)_{mm} = Q^m$ and*

$$\left|(\widehat{R}_1)_{m,m-1}\right| = \sqrt{(1-Q^m)(Q^{m-1}-B^{-1})} \le \sqrt{1-Q^m} \le \sqrt{m\theta}.$$

Proof. By Proposition 5.2,

$$\frac{D_m}{D_{m-1}} = -\frac{BQ^{2m-2}(1-Q^m)}{BQ^{m-1}-1}, \qquad (\widehat{R}_1)_{m,m-1} = (R_1)_{m,m-1}\,\frac{|D_{m-1}|^{1/2}}{|D_m|^{1/2}}.$$

Squaring and simplifying gives the identity; then use $0 < Q^{m-1} - B^{-1} < 1$ and $1 - e^{-m\theta} \le m\theta$. □

Remark 6.3. For the q-Vandermonde reference of [10] the corresponding quantity is $\sqrt{1-Q^m}$; the new reference multiplies it by $\sqrt{Q^{m-1}-B^{-1}} \approx \sqrt{1-1/\alpha} < 1$. Nothing else in the dilation argument depends on the pivots.

Lemma 6.4 (Generator)**.** *Let $x = r\theta \le 1/64$. Then $N := \widehat{R}_1 - I$ satisfies $\|N\| \le x + \sqrt{x} \le 9/64$, the principal logarithm $G = \log \widehat{R}_1 = \sum_{j\ge1} (-1)^{j+1} N^j/j$ satisfies $\|G\| \le \frac{72}{55}\sqrt{x}$, $\widehat{R}_\beta = e^{\beta G}$ for all real β, and*

$$\theta d + 2\|G\| \le 4\sqrt{r\theta} =: t.$$

Proof. The diagonal part of N has entries $|Q^m - 1| \le m\theta \le x$ and its sub-diagonal part has entries $\le \sqrt{x}$ (Lemma 6.2); a diagonal and a weighted shift have norms equal to their largest entries, so $\|N\| \le x + \sqrt{x}$. Then $\|G\| \le -\log(1-\|N\|) \le \|N\|/(1-\|N\|) \le \frac{64}{55}\cdot\frac{9}{8}\sqrt{x}$. Next, $R_\beta = \exp(\beta L^{-1}\Lambda' L)$ with $\Lambda' = \mathrm{diag}(i \log Q)$, so $\widehat{R}_\beta = e^{\beta \widehat{G}}$ with $\widehat{G} = |D|^{-1/2} L^{-1} \Lambda' L |D|^{1/2}$. Both $\widehat{G}$ and G equal $P(\widehat{R}_1)$, where P is the Lagrange polynomial with $P(Q^i) = i \log Q$ (the eigenvalues Q^i of $\widehat{R}_1$ are distinct and $\widehat{R}_1$ is diagonalizable); hence $G = \widehat{G}$. Finally $\theta d + 2\|G\| \le x + \frac{144}{55}\sqrt{x} \le (\frac{1}{8} + \frac{144}{55})\sqrt{x} < 4\sqrt{x}$. □

Proposition 6.5 (All-degree response)**.** *If $r\theta \le 1/64$, then $\mathcal{R}\big((\theta s)^n\big) \le t^n$ for every integer $n \ge 0$, with $t = 4\sqrt{r\theta}$.*

Proof. For $s = d - i - j$ we have $e^{\beta\theta s} = e^{\beta\theta d} Q^{\beta i} Q^{\beta j}$, so $M \circ e^{\beta\theta s} = e^{\beta\theta d} \Lambda_\beta M \Lambda_\beta$. Since A commutes with Λ_β, $S^{-1}\Lambda_\beta S = |D|^{-1/2} L^{-1} \Lambda_\beta L |D|^{1/2} = \widehat{R}_\beta = e^{\beta G}$, and therefore

$$F(\beta) := S^{-1}\big(M \circ e^{\beta\theta s}\big) S^{-T} = e^{\beta\theta d}\, e^{\beta G}\, J\, e^{\beta G^T}.$$

Entrywise, $F^{(n)}(0) = S^{-1}(M \circ (\theta s)^n) S^{-T}$. Expanding the three exponentials, $F^{(n)}(0) = \sum_{a+b+c=n} \frac{n!}{a!\,b!\,c!} (\theta d)^a G^b J (G^T)^c$, whose norm is at most $(\theta d + 2\|G\|)^n \le t^n$. □

Definition 6.6 (Weighted algebra)**.** For $w \geq 1$ let $\mathcal{A}_w$ be the space of power series $\phi(s) = \sum_{n\geq 0} \phi_n s^n$ with $\|\phi\|_{\mathcal{A}} = \sum_n |\phi_n| w^n < \infty$. It is a commutative Banach algebra, and $\|e^{\phi} - 1\|_{\mathcal{A}} \leq e^{\|\phi\|_{\mathcal{A}}} - 1$.

Corollary 6.7. *Let $r\theta \leq 1/64$ and $w = t/\theta = 4\sqrt{r/\theta}$. Then $w \geq 32r > d$, and for every $\phi \in \mathcal{A}_w$, $\mathcal{R}(\phi) \leq \|\phi\|_{\mathcal{A}}$.*

Proof. $w/r = 4/\sqrt{r\theta} \geq 32$. The series $\phi(s) = \sum \phi_n s^n$ converges absolutely at every integer $|s| \leq d$, so $M \circ \phi = \sum_n \phi_n\, M \circ s^n$ entrywise, and $\mathcal{R}(s^n) = \theta^{-n}\mathcal{R}((\theta s)^n) \leq w^n$. □

7. The four-zero remainder

Let $k \geq 10^9$ and $r\theta \leq 1/64$. By Input A, $\log b_s = f(k+s) - f(k) - s\,(f(k+1) - f(k))$ at integers $|s| \leq d$, and the right-hand side is holomorphic for $|s| < 0.05k$. Define

$$h(s) = f(k+s) - f(k) - s\,(f(k+1) - f(k)) - \ell(s), \qquad \beta_n = [s^n]h(s).$$

Then $b_s = c_s\, e^{h(s)}$ at the integers $|s| \leq d$, and, by Lemma 4.2,

$$h(-1) = h(0) = h(1) = h(2) = 0.$$

Lemma 7.1. *For $n \geq 3$, $|\beta_n| \leq 3 \cdot 40^n k^{1-n} + (9/k)^{n-1} \leq 4 \cdot 40^n k^{1-n}$.*

Proof. For $n \geq 2$, $\beta_n = f^{(n)}(k)/n! - [s^n]\ell$; use (2.1), Lemma 4.6 and $9^{n-1} \leq 40^n$. □

Let $\mathcal{I}_3$ denote Lagrange interpolation at the nodes $-1, 0, 1, 2$, with basis

$$\ell_{-1}(s) = -\tfrac{s^3}{6} + \tfrac{s^2}{2} - \tfrac{s}{3}, \quad \ell_0(s) = \tfrac{(s+1)(s-1)(s-2)}{2}, \quad \ell_1(s) = -\tfrac{s^3}{2} + \tfrac{s^2}{2} + s, \quad \ell_2(s) = \tfrac{s^3}{6} - \tfrac{s}{6},$$

and put $P_n(s) = s^n - \mathcal{I}_3[s^n]$; note $P_n = 0$ for $n \leq 3$.

Lemma 7.2. *If $w \geq 1$ and $n \geq 4$, then $\|\mathcal{I}_3[s^n]\|_{\mathcal{A}} \leq 2^n w^3$ and $\|P_n\|_{\mathcal{A}} \leq w^n + 2^n w^3$.*

Proof. For $n \geq 1$, $\mathcal{I}_3[s^n] = (-1)^n \ell_{-1} + \ell_1 + 2^n \ell_2$. For $w \geq 1$, $\|\ell_{-1}\|_{\mathcal{A}} \leq w^3$, $\|\ell_1\|_{\mathcal{A}} \leq 2w^3$, $\|\ell_2\|_{\mathcal{A}} \leq w^3/3$, and $3 + 2^n/3 \leq 2^n$ for $n \geq 3$. □

Proposition 7.3. *Assume $k \geq 10^9$, $r\theta \leq 1/64$, $w = 4\sqrt{r/\theta}$ and $y := 40w/k < 1$. Then*

$$h = \sum_{n\geq 4} \beta_n P_n \quad \text{in } \mathcal{A}_w, \qquad \|h\|_{\mathcal{A}} \leq 4k\,\frac{y^4}{1-y} + 4k\,w^3\,\frac{(80/k)^4}{1 - 80/k}.$$

Proof. By Lemmas 7.1 and 7.2, the series $\sum_{n\geq 4} \beta_n P_n$ converges absolutely in $\mathcal{A}_w$, and $\sum_n \beta_n \mathcal{I}_3[s^n]$ converges to a cubic polynomial. Hence

$$p := h - \sum_{n\geq 4} \beta_n P_n = \sum_{n\leq 3} \beta_n s^n + \sum_{n\geq 4} \beta_n \mathcal{I}_3[s^n]$$

is a cubic polynomial; it vanishes at the four nodes because h and every P_n do, so $p \equiv 0$. The bound follows by summing $4 \cdot 40^n k^{1-n}(w^n + 2^n w^3)$ over $n \geq 4$. □

The first term is the quartic main term: since $w^2 = 16r/\theta$ and $k\theta \asymp 1$, it is $\asymp k^{-3}w^4 \asymp r^2/k$. For the three-point reference the analogous main term is $\asymp k^{-2}w^3 \asymp (r^3/k)^{1/2}$, which is the source of the cubic wedge (1.2).

8. Proof of the Main Theorem

Lemma 8.1 (Inertia)**.** *Let S be invertible, J a signature matrix, and $T = S(J+F)S^T$ with F symmetric and $\|F\| < 1$. Then T and J have the same inertia.*

Proof. For $\lambda \in [0,1]$, $\sigma_{\min}(J+\lambda F) \geq 1 - \lambda\|F\| > 0$, so the eigenvalues of the symmetric family $J + \lambda F$ never cross 0; hence $J + F$ has the inertia of J, and Sylvester's law of inertia transfers it to T. □

Proof of the Main Theorem. Let $r \geq 2$ and $k \geq 10^{11} r^2$; then $k \geq 4 \cdot 10^{11}$, so Inputs A and B and all lemmas above apply. By Lemma 4.3, $r\theta < 2.002\, r/k \leq 2.002/(10^{11} r)$, which is $\leq 1/64$ and $< \log 1.45 < \log \alpha$; so Lemma 4.4, (5.1), Corollary 6.7 and Proposition 7.3 hold. Since $k\theta > 0.53$, $w^2 = 16r/\theta < 16rk/0.53$ and

$$y^2 = \frac{1600\, w^2}{k^2} < \frac{25600}{0.53} \cdot \frac{r}{k} \leq \frac{25600}{0.53 \cdot 2 \cdot 10^{11}} < 2.5 \cdot 10^{-7}.$$

Hence $4ky^4/(1-y) < 4\,(25600/0.53)^2\, 10^{-11}/(1-y) < 0.09337$, and the second term in Proposition 7.3 equals $4 \cdot 80^4 (y/40)^3/(1 - 80/k) < 3.1 \cdot 10^{-7}$. Therefore $\|h\|_{\mathcal{A}} < 0.0934$ and $\|e^h - 1\|_{\mathcal{A}} < e^{0.0934} - 1 < 0.098$ (Appendix A).

Let $H' = [b_{d-i-j}]_{i,j=0}^{d}$. Since $b_s = c_s e^{h(s)}$ at the integers $|s| \leq d$, $H' - M = M \circ (e^h - 1)$, and Corollary 6.7 gives $\|S^{-1}(H'-M)S^{-T}\| \leq \|e^h - 1\|_{\mathcal{A}} < 1$. By (5.1) and Lemma 8.1, H' has the inertia of J; J has $\lfloor r/2 \rfloor$ negative entries, so $\operatorname{sgn}\det H' = (-1)^{\lfloor r/2 \rfloor}$.

Finally, $a_{k+d-i-j} = a_k \rho^d\, \rho^{-i} \rho^{-j}\, b_{d-i-j}$ (all indices are $\geq k - d > 0$), so $H = [a_{k+d-i-j}]$ is a positive diagonal congruence of a positive multiple of H', and $\operatorname{sgn}\det H = (-1)^{\lfloor r/2 \rfloor}$. Reversing the order of the columns of $[a_{k+j-i}]$ gives H, at the cost of the sign of the reversal permutation, $(-1)^{\lfloor r/2 \rfloor}$. Hence $D_{r,k} = (-1)^{\lfloor r/2 \rfloor} \det H > 0$. □

Remark 8.2 (The constant). The main term is at most $4(25600/0.53)^2/C$ when $k \geq Cr^2$, so the argument closes as soon as this is $< \log 2$, i.e. for every $C > 1.35 \cdot 10^{10}$. Using the cruder windows $k\theta > 0.4$, $\alpha > 4/3$ one gets $C > 2.37 \cdot 10^{10}$. No effort has been made to optimize Input A, which is the dominant loss (the true coefficients satisfy $|f^{(n)}(k)/n!|\, k^{n-1} < 0.3$, cf. Table 2, versus $3 \cdot 40^n$).

9. Numerical illustration

Nothing in this section is used in the proof. We use the certified values of λ_n, $n \leq 12016$, published with [9] (relative width $< 3 \cdot 10^{-91}$), which we cross-checked against our own quadrature of (1.1). At $k = 12000$ one finds $k\tau_k = 1.74995$, $k^2|\tau_k - \mathcal{T}(k)| = 1.017$ (compare 20 in Input B), $k^2 \Delta_k = 1.7225$, $\alpha = 1.7777$, $k\theta = 0.7655$.

Remark 9.1 (Outlook). Heuristically, a reference with m exact matching points and an exact LDL^T with q-Pascal factor would give a main term $\asymp k^{1-m} w^m \asymp r^{m/2} k^{1-m/2}$, i.e. a wedge $k \gtrsim r^{m/(m-2)}$. Whether such references exist for $m \geq 5$ (for instance among ${}_r\phi_s$-type sequences) is open; in any case this route cannot reach the critical regime $k = O(r)$, where w becomes comparable to the radius of analyticity.

k	r	three-point	four-point	ratio	four-point$\cdot k/r^2$
11990	3	$4.86 \cdot 10^{-3}$	$1.77 \cdot 10^{-5}$	275	0.024
11990	6	$4.00 \cdot 10^{-2}$	$2.17 \cdot 10^{-4}$	184	0.072
11990	8	$7.93 \cdot 10^{-2}$	$5.02 \cdot 10^{-4}$	158	0.094
11990	12	$1.88 \cdot 10^{-1}$	$1.49 \cdot 10^{-3}$	127	0.124
11990	16	$3.31 \cdot 10^{-1}$	$3.06 \cdot 10^{-3}$	108	0.144
3000	8	$1.61 \cdot 10^{-1}$	$1.93 \cdot 10^{-3}$	83	0.091
6000	8	$1.13 \cdot 10^{-1}$	$9.86 \cdot 10^{-4}$	114	0.092

Table 1: Actual whitened perturbation $\|S^{-1}(H' - M)S^{-T}\|$ on the true ξ-coefficients, for the three-point reference of [10] and the four-point reference of this paper. At fixed r the four-point error decays like k^{-1} and the three-point error like $k^{-1/2}$.

n	$\lvert f^{(n)}(k)/n!\rvert\, k^{n-1}$	$\lvert[s^n]\ell\rvert\, k^{n-1}$	$\lvert\beta_n\rvert\, k^{n-1}$	bound $4 \cdot 40^n$
3	0.287	0.287	$9.0 \cdot 10^{-6}$	$2.6 \cdot 10^5$
4	0.142	0.196	0.054	$1.0 \cdot 10^7$
5	0.085	0.157	0.072	$4.1 \cdot 10^8$
6	0.056	0.139	0.083	$1.6 \cdot 10^{10}$
8	0.030	0.132	0.102	$2.6 \cdot 10^{13}$

Table 2: Taylor coefficients at $k = 12000$, computed by Cauchy integrals of $\log I(z) - \log\Gamma(2z+1)$ and of (4.1). The predicted size of the dominant term $\beta_4\mathcal{R}(P_4)$ at $(k, r) = (11990, 8)$ is $5.0 \cdot 10^{-4}$, in agreement with Table 1.

A. Certified scalar inequalities and reproducibility

All scalar inequalities in Sections 3–8 are verified by the ancillary script `certificate.py` using Arb ball arithmetic [7] at 256 bits. Each window has the form $c(k)/k$ or $c(k)/k^2$ with c monotone in the favourable direction, so evaluation at $k = 12000$ suffices. The certified values are:

quantity	certified enclosure	used
$k\tau_k$	$[1.712619,\ 2.001667]$	(1.71, 2.002)
$k^2\Delta_k$	$[1.675686,\ 2.003334]$	(1.675, 2.004)
α	$[1.458989,\ 2.392839]$	(1.45, 2.40)
$k\theta$	$[0.537774,\ 2.002]$	(0.53, 2.002)
$k\kappa$	≤ 6.451	< 8.4
y (wedge)	$\le 4.915 \cdot 10^{-4}$	
$\|h\|_{\mathcal{A}}$ (wedge)	≤ 0.0933691	< 0.0934
$\|e^h - 1\|_{\mathcal{A}}$	< 0.098	< 1

Further ancillary scripts (not needed for the proof) check: `sym_check.py` — symbolic verification (generic parameters) of Lemma 4.2, the integer cocycle, Lemma 5.1, Proposition 5.2 for $i, j < 7$, Lemmas 6.1–6.2 and Lemma 7.2; `gateB_numeric.py`, `response_scaling.py` — the response bound of Proposition 6.5, computed from a numerical LDL^T of M at 260 digits; `xi_data.py`, `whitened_experiment.py`, `taylor_check.py` — Section 9.

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Use of AI tools

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References

[1] M. Aissen, I. J. Schoenberg, A. M. Whitney, *On the generating functions of totally positive sequences I*, J. Analyse Math. **2** (1952), 93–103.

[2] G. Csordas, T. S. Norfolk, R. S. Varga, *The Riemann hypothesis and the Turán inequalities*, Trans. Amer. Math. Soc. **296** (1986), 521–541, doi:10.2307/2000378.

[3] A. Edrei, *On the generating functions of totally positive sequences II*, J. Analyse Math. **2** (1952), doi:10.1007/BF02786971.

[4] G. Gasper, M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Cambridge University Press, 2004.

[5] M. Griffin, K. Ono, L. Rolen, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), no. 23, 11103–11110.

[6] J. Holland, *A new hyperbolicity wedge and a joint semicircle limit for Jensen polynomials of Riemann's ξ-function*, arXiv:2608.08682 (2026).

[7] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292.

[8] O. M. Katkova, *Multiple positivity and the Riemann zeta-function*, Comput. Methods Funct. Theory **7** (2007), no. 1, 13–31; arXiv:math/0505174.

[9] Y. Liu, J. Mao, *Infinite log-concavity of the Taylor coefficients of the Riemann xi-function*, arXiv:2610.04972 (2026).

[10] W. Michałowski, *An explicit uniform cubic wedge for consecutive Toeplitz minors of the Riemann xi coefficients*, arXiv:2607.16795 (2026).

[11] J. Nuttall, *Wronskians, cumulants, and the Riemann hypothesis*, Constr. Approx. **38** (2013), 193–212, doi:10.1007/s00365-013-9200-8.

[12] S. Sra, *On some positivity properties of the Riemann Ξ function*, arXiv:2611.02944 (2026).

[13] L. X. W. Wang, N. N. Y. Yang, *Laguerre inequalities and complete monotonicity for the Riemann Xi-function and the partition function*, Trans. Amer. Math. Soc. (2024).